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Traceability and Accounts

Overview

In October 2022 one of my sons ran the London marathon, thirty-two years after my own effort on the same course. In contrast to 1990, he was able to track his progress on a digital map of the London route via an app linked to his digital watch. This tracking application analysed his run in terms of "splits" and produced an overall summary of his "performance" beyond simply the time taken. The app also enabled him to monitor his heart rate. He became concerned at its level at the 32 kilometre stage and decided to slow down. In short, like many other runners that day, my son was "accounting" for himself in real time; he was able to self-monitor as I never could. Tracking his progress from a distance, using the same app, there was also a scare when his icon was stationary for a considerable length of time, only suddenly to jump to the finish. Had he "beamed" himself to the end? Perhaps that smart watch was smarter than I could ever imagine? But in fact the app connectivity had temporarily failed; the experience of tracking his progress in so-called real time was imperfect, with delays, lags, and correcting jumps. Clearly, much more is needed than a smart watch attached to a runner's body for successful tracking. An entire background apparatus must be in place, involving programmers, satellites, manufacturers, internet providers, and many more agencies, all distributed but also all connected and somehow interdependent. From this point of view, what becomes surprising is that the app worked at all—not that it misfired a few times, no doubt due to the volume of use that day.

Vignettes like this, which point to the entanglement of digital devices in our everyday lives, are increasingly familiar and well documented by scholars. This one has a number of ingredients worthy of note. First, there is the technology-enabled *traceification* of human action by which (some) bodily movements and states are captured and marked by a machine. Second, this traceification process enables the *tracking* of the runner both by himself and also by remote observers using the app, thereby increasing their observational reach. Third, traceification is also generative of a digital double or proxy of the runner, defined by various biometrics that provide evidence of "how he is doing". In other words, the generated traces of this app support *accounts* of his performance and bodily functions. The vignette is an example of what has been called the "culture of self-tracking" in which humans hold themselves to account and audit themselves (Lupton, 2016b). In the background, there is necessarily an infrastructure or platform that produces the app, provides the necessary *connectivity* for it to work, and which may itself co-opt the traces produce by the runner and datafy them for future commercial benefit.

Here then are most of the conceptual elements for the arguments in this book: traces, traceification, datafication, tracking, accounting, and connectivity. Of particular note, the vignette implies a tight relationship between the production of traces and their conversion into the performance data that make up accounts. The analysis of this relationship and its dynamics is a core focus for this book, and the ambition is to develop a *traceological theorization of accounting* that can be distinguished from an information perspective on decision-making, on the one hand, and from a sociology of quantification perspective on reduction and commensurability, on the other.

Remarkable things in our everyday lives, such as remotely tracking a relative doing a charity run, often become unremarkable to us. It is one of the tasks of social science, and philosophy, to recover that sense of the remarkable, and to problematize and to investigate the conditions of possibility for the production of the fabric of our everyday world. But such an analytic attitude is also methodologically daunting. In this book we endeavour to look through and beyond this specific and personal vignette to disclose aspects of the emerging form of life of which it is symptomatic. As we do so, we see that something bigger and more systematic is going on. It may seem at first glance that there is nothing to link this use of a smart watch during a marathon to, say, wine labelling, anti-money-laundering regulation, financial accounting, the legalization of cannabis, cryptocurrency trading, fingerprinting, credit scoring, and food-quality certification. Each of these practices and fields, and many more besides, has its own apparatuses of devices and knowledge production. Each has its own specialist followers and observers in industry and academic circles, with learned journals dedicated to the steady elaboration of a science of its practice. And each is mobilized, it seems, by different values and goals, focused on different objects. Indeed, we should not be surprised by this diversity; it simply reflects how our world is a complex portfolio of endeavours addressing distinct social and economic concerns and values.

Yet, these many diverse worlds of practice share a feature with each other, and with the smart watch example, which might seem unremarkable, but which points to a family resemblance worthy of further reflection. As we analyse in greater depth in this book, they each exhibit the ingredients already detected in the vignette, ingredients that combine to constitute practices of *traceability*. They are each involved with making specific things, processes, and people knowable, visible, and connected via the traces that they produce or that are produced of them, which they leave behind or which follow them, and which enable them to be tracked (i.e. to be observable and controllable).

By construction, traceability necessarily begins with traceification, with making things traceable by generating traces of them. Yet, traces are empirically heterogeneous. Some, like labels and tags, move with the objects and people to which they are attached, providing evidence of origins or quality, a process that may be more or less intensively regulated. Some, as in the marathon runner case, may be produced by sensors attached to human bodies, to enable the monitoring of blood pressure, distance run, calories eaten, emotional state, sleep quality, and much more. Some traces may take the form of routinely produced documentary inscriptions, records of processes and decisions, such as checklists and minutes. And some may be the results of scientific enquiry using laboratory instruments to identify, isolate, and even in some cases generate them under controlled conditions. In many of these instances, traces are produced, accumulate, and, with work, become organized as data to support statements of fact and meaningful aggregated accounts of personal and organizational performance.

Beyond this descriptive snapshot of both the extent and variability of the settings of traceability, why should we care about it? The thesis developed in this book is that we are experiencing a systemic shift in the nature of traceability and its components, a shift with profound consequences for human governance and oversight, for "the accounts we live by", for the nature of our experience and selfhood, and even for the conduct of the social sciences. To take one theme that is developed further in several chapters, digital transformations in all their variety are challenging the "audit society" model and its underlying hierarchical mode of governance based on the logic of the audit trail (Power, 1997, 2021a). These traditional oversight structures and their forms of accounting are being systematically transformed and, in a manner to be explained, "flattened" and "delayered". More generally, the purpose of this book is to understand the nature and potential consequences of this systemic transition from one economy of traces, based on the "documentality" of the audit trail, to another based on digital "traceification" and tracking. A traceological approach to accounting is developed that enables us to theorize this transition and its tensions.

To execute the theoretical task, it is necessary to be not only mindful of the power of digital technologies but also conceptually distant from them. Such an approach is risky, not least because it necessarily cuts across a formidable body of detailed work by scholars in information systems, surveillance studies, and the sociology of technology, to name only a few. We therefore begin this project in the next section with some general reflections on digitization and its implications for the organization of traceability in an economy of traces. Following these positional reflections, we provide general introductions to both traceability and accounting. We analyse the idea of traceability as it has become a value and purpose, what we call a *value-purpose*, for states, organizations, and individuals. Three broad value-purposes are proposed and elaborated—quality, security, and discovery—in order to provide further empirical texture to the landscape of practices. We then develop and justify a broad notion of accounting that is more than the production of information for decision-making and which is significant for the constitution of our individual, organizational, and collective lives, shaping experience, acts of organizing, and the very nature of what it is to "perform" (Lupton, 2016b, 2016c; Vormbusch, 2022). This broad conception draws upon, and seeks to recover, the continuities between everyday forms of "interactive" account-giving (Garfinkel, 1984a [1967]; Butler, 2005) and formal organizational accounting (Jeacle and Carter, 2023; Vollmer, 2019), continuities that have become more evident in the digital age. We use the terms "accounts" and "accounting" interchangeably to express these continuities and exchanges.

Finally, we introduce the individual chapters of the book and their respective thematic emphases, which will deepen these introductory remarks. The aspiration is to theorize traceability in a way that is "digitally neutral" but also mindful of, and engaged with, the implications of digital developments in a general sense. In this way, the analysis seeks to avoid being hooked by the "euphorias and paranoias of speed and change" (Haraway, quoted in Gane, 2006, p. 155) in order to understand how the economy of traces is shifting and what is at stake.

Economy of Traces

Scholars develop motifs and labels to dramatize new forms of social ordering. This is often how theorizing begins (Swedberg, 2014), and we are familiar with many of them: "algorithmic society" (Schuilenburg and Peeters, 2020); "granuläre Gesellschaft" (Kucklick, 2014); "embodied computing" (Pedersen and Iliadis, 2020); "surveillance society" (Lyon, 2001); "surveillance capitalism" (Zuboff, 2019); "risk society" (Beck, 1992); and "audit society" (Power, 1997), to name a few. The "economy of traces" is no less an organizing and theoretical motif of this kind. It is a critical notion originally coined by Stiegler (2016, p. 23) to characterize an increasingly pervasive and reductive automation that constitutes the enabling infrastructure for a new form of dehumanizing capitalism and assetization. In this book, we co-opt Stiegler's idea and use it to reference traceability, understood as a systemic, powerful, value-laden, but often hidden cultural logic of trace-production, organizing and accounting for ourselves both individually and collectively.

Within this economy of traces, personal tracking devices and wearables like watches with biosensors have not only become feeders for what Zuboff calls surveillance assets, but more generally, as we argue in Chapter 8, they also expand the realm of performing and performance. Even where such performances do not involve direct forms of monetary exchange or rewards, and even when gamified, they represent a form of "economization" in the sense of generating an orientation towards the metrics of performing. Going back to the vignette with which this chapter began, it is plausible to say that no human would have imagined using a watch in this way until it was possible and marketed to them. Yet it is also undoubtedly the case that cultures of self-monitoring and self-improvement have always existed. From the Socratic imperative to "know thyself", to ancient manuals of self-help, proper statehood, and commercial conduct, to the volumes that are easily found today in airport bookshops, preoccupations with self-monitoring existed before the explosion of biosensing wearables. In other words, successful digital technologies have been able to draw upon, supercharge, and economize pre-existing cultural dispositions to self-monitor. In this way, the economy of traces is both epistemic, as a form of knowing things and people, and political and moral in its implications.

Undeniably, these changes also represent business transformations amenable to economic analysis, for which platforms signify novel forms of economic organization. However, in the arguments that follow, we steer towards an infrastructural sense of the "economy of traces" as shorthand for the cultural organization of traceability. In effect, the motif of "economy of traces" references the sum of all forms of organized traceability practices and their enabling features of traceification, datafication, tracking, connecting, and accounting.

Concerns with traceability are long-standing and widespread, permeating many fields of practice, but they are also complex and variable. For example, practices of tracking may be animated by ideals of transparency and demands for evidence, but equally they can be objects of critique in the face of threats to privacy (Bernstein, 2012). The nature and extent of traceification also vary considerably according to the technological capacities that enable connectivity, tracing, and tracking (Leonardi and Treem, 2020; Flyverbom, 2021). And it is clear to any casual observer of our contemporary world that the expanded and expanding reach of automation, in the form of various digital technologies and devices, has supercharged and intensified possibilities for traceability, for generating new relations and objects and especially new forms of sociality (Pacchera et al., 2023). Digital traces enable practices of tracking that are now faster, less intermediated, and more granular in form, ranging widely from the use of barcodes, to GPS tracking, to blockchains, and now machine learning (Thylstrup, Archer, and Ravn, 2022). This expanded capability for trace production and hence for the tracking of things and people is not confined to the digital transformation of the routines and work processes of states and organizations. It has also become a prominent feature of private and personal activity, supported by a wide variety of social media platforms and devices that humans willingly attach to their bodies (Lupton, 2016a; Vormbusch, 2022). Indeed, many scholars and policymakers now agree that societies stand at the threshold of a new future, in which the digitally enabled automation of everything, including vast areas of human decision-making being displaced by machine-learning, has become not only possible but actual (Flyverbom, 2022; Zuboff, 2022).

These developments are known to be "disruptive" for traditional business models. They promise the disintermediation of many transaction services by placing control directly in the hands of consumers. They also challenge the very notion of an "organization" with well-defined boundaries and are argued to be generative and creative. It is not just a matter of "automation"; they are formative of newly visible objects, capabilities, and connectivities (Mennicken and Kornberger, 2021). Digital technologies also have profound ethical implications for how we think of, and account for, ourselves, for the nature of privacy and intimacy, and for understanding the place of humans in these newly constituted social and economic practices (Boellstorff, 2016; Velliz, 2023). To adapt an idiom from the philosopher Ian Hacking, it seems as if we are experiencing an "avalanche of traces" with far-reaching potential to redefine what it is to be human, what it is to "perform" across many diverse areas of life, and even what it is to be real (Andrejevic, 2013; Borgmann, 1999). Critical voices are concerned with how human physical actions are being captured as "clicks," stored and operated upon by machines and clustered into new objects and categories by powerful algorithms (Zuboff, 2019, 2022). Such algorithms, understood as codified instructions for processing traces of the human, are also at the heart of what scholars now called "datafication" (Alaimo and Kallinikos, 2024).

While many celebrate the great benefits and augmentative power of these developments, others fear for the future of human experience as we become "hooked" by, and on, data (Seaver, 2019), even to the point of becoming hybrids of digital data and human flesh (Gane, 2006:p. 2). Privacy, science, and democracy are argued to be at risk, and new kinds of discrimination are emerging from powerful apparatuses of surveillance fuelled by the expansion of artificial intelligence (AI) (Velliz, 2023). Furthermore, long-standing, juridically grounded forms of accountability and governance in many societies are being fundamentally reconstructed, and digitally enabled traceability in many areas is now big business as consumers and state apparatuses demand greater connectivity and efficiency in their quests for secure transacting free of risky intermediaries. There is also a desire to know more about the origins and reliability of things, especially food, and people. As this demand for traceability as evidentiary demonstrability, and the digital capability to address that demand, continue to co-evolve, critics like Stiegler (2016, p. 14) suggest that "we are witnessing the establishment of that complete and general automatization made possible by the industry of reticulated digital traces". He suggests that we now live in a pervasive non-documental "economy of traces" (pp. 22–27) whose formative logic we are only beginning to understand.

Digital transformations over at least three decades have undoubtedly accelerated and expanded the possibilities for practices of tracing and tracking. For example, advances in chip technology have enabled information processing to increase capacity, speed, and granularity. These functional attributes have come to be highly valued by organizations, and positive discourses of the augmentative power of digital technologies for individuals, businesses, and medical practice are widespread. However, as information systems scholars and historians of technology know well, what is loosely referred to as the "digital revolution" is not a single thing and certainly not a single kind of event. There are overlapping and temporal sequences with different emphases (Kolb et al., 2020).

From the invention of the internet and barcodes to blockchain, big data, and AI; from mainframes to personal computing and smart phones; from databases to Enterprise Resource Planning (ERP) systems and data warehouses, there is commonality. Indeed, all are based in some sense on the conversion of physical and material traces into mainly binary electronic pulses and then into combinable forms of notation (Kitchin, 2022; Alaimo and Kallinikos, 2024, ch. 3). There have also been profound underlying transformations in processing power, speed, and storage capacity underlying an ongoing transition from the digital automation of existing analog processes to forms of machine learning powered by meta-algorithms, which adapt to their experiences and produce new algorithms for data analysis and accounts production. Puranam (2025) conceptualizes digital algorithmic transformation as having three basic stages: digital representation (datafication); pooling (volume); and aggregation (valuable patterns for potential prediction). Indeed, these three stages are generic and apply to conventional accounting practices in which transactions are recognized and recorded, pooled by double-entry bookkeeping systems and aggregated into meaningful accounts.

Yet, beyond these commonalities there are also very different technologies and devices in play with different histories and maturities, different user communities, different applications and aspirations, and therefore different implications for the shape and scope of traceability potential and practice. Furthermore, the hegemony that critics often attribute to some, or even all, of these developments should be handled with caution. There is a danger of both over- and understating the dystopian risks identified by writers such as Zuboff (2019), Stiegler (2016), and many others. For example, Alaimo and Kallinikos (2024, p. 99) argue that we need to reach beyond macro-conceptual and negative framings of surveillance to understand its many forms and generative properties. Furthermore, despite societal demands for regulation and control, tracking and monitoring practices cannot easily be disentangled from the infrastructures in which they are embedded. This book aims at a theoretical analysis that is mindful of these complex and multifaceted developments and does not ignore them, but which is also, initially at least, agnostic about their newness and impact, even as we become "born digital", in order to avoid the dual extremes of celebration of the automated augmentation of human tracing capacity, on the one hand, and humanist blanket critique, on the other. Indeed, while the contemporary importance of digital technologies for trace production and expanding processes of datafication and tracking cannot be underestimated, and the rise of AI does seem to be a fundamental rupture, they are also long-standing features of calculative human practices (Lanzolla et al., 2020). Before we are intellectually swamped by the seeming digitization of everything, in which traces of nearly every conceivable human action can be created and processed, there remains a place for reflection on that which is continuous with earlier documentary practices. Here we aim at theorizing that is as "digitally neutral" as possible, while being nevertheless cognizant of the enormous impact of advances in digital technologies.

There are good theoretical reasons for attempting this digitally agnostic approach. The "digital" is itself a problematic category, for both historical and semiotic reasons (e.g. Haugeland, 1981). Scholars have shown that much more is at stake than just the rendering of information into computer-machine readable form, and there have been notable efforts to provide working taxonomies. According to Leonardi and Treem (2020, p. 1605), "digitization" refers to "the encoding of actions or representations of actions into a digital format (zeros and ones) that can be read, processed, transmitted and stored by computational technologies". "Digitalization" refers to "The ways in which social life is organized through and around digital technologies". "Datafication" refers to "The practice of taking an activity, behaviour, or process and turning it into meaningful data". Yet, as heuristically helpful as this and other taxonomies and their distinctions may be, they are often embroiled in each other and are interdependent: digitization preloads datafication, and processes of digitalization will influence practices of action-encoding or traceification. Unless otherwise necessary, we will tend to use the term "digitalization" to emphasize the embedded nature of digital technologies and the sense in which organizational work practices and technology co-evolve.

Another reason to avoid a fully digitally conversant entry point to the study of traceability is that the digital, understood more broadly than its computer-electronics setting, is a logic of (binary) distinctions and discrete units, and its opposite—the analogic—is more to be understood as process, continuity, and messiness. Both these orientations have been around a long time, and the encoding of actions in durable form—in what we call inscriptions—existed long before many of the developments discussed under the contemporary notion of digitization. Furthermore, the so-called "digital age" did not arrive suddenly, although there have been significant moments of change, not least the invention of the World Wide Web, the rise of home computer power, and the arrival of the smart phone. Alaimo and Kallinikos (2024, ch. 3) show how data emerged from recording technologies such as the punch card, whose material perforations progressively became autonomous semiotic tokens in a larger epistemic and economic system. Debates about the effects of automation on labour and the de-skilling problem also have a long history.

Galloway (2022, p. 228) argues that "digitality and analogicity are general modes of mediation; they are not facts about consumer electronics". Digitization should be understood more broadly as a process of making things discrete, and of finding notation to do that (Shannon, 1948), whereas analogicism is an orientation to synthetic qualities of liquidity and flow. From this larger point of view, it is somewhat ironic that the datafication phase of the digital revolution has occurred at a time when so many areas of organization studies, humanities, social sciences, and intellectual life in general, including the underlying theoretical orientation of this book, embrace forms of pragmatism, processualism, emergentism, relationality, liquidity, and heterogeneity—exactly the characteristics attributed to analog complexes. In this sense we also live now in a golden age of analog, according to Galloway, which is not to be read as "old" or "offline" but as a mode of mediation that aims at being expansive and non-reductive.

Daston's (2022) history of the rules we live by draws attention to another category—that of the "algorithm" which preoccupies contemporary discussion. Algorithms were also systems of rules that existed prior to any machine dreams of automation. The "thinning out" of rules in mathematical algorithms and their embodiment in machines for mechanical calculation became an early model of organizing as such. Taylorism and its variants are in this sense algorithmic models of organization, directed at eliminating the contextuality of the rules by which work is performed. Practices of quantification, including accounting, were always algorithmic and "digital" in this broader sense of being reductive, commensurating, and binary in their logic—we need only think of the debit and credit in double-entry bookkeeping. Hence, many algorithmic practices were always in effect pre-automated and "in waiting" for the computing capability to complete the automation of their binary mode of mediating.

The advantage of making the effort to step back from conventional notions of the digital and digitization is that it allows something else important to come into view, namely that traces can be *both* analog and digital in nature. Analogically they can be conceptualized as idiosyncratic signs and cues that support pragmatic and vernacular knowledges of weather, other people, the Will of God, and so on. Understood as discrete, homogenous, or commensurable units based on a duality of signifier and signified, traces can also be "digital" and are more closely associated with the idea of data. Traces in the coded form of binary electronic pulses are just a special case of the digital in its broader sense of discrete-ism and granularity, and algorithms are the operating codes that use them as input.

While we should not detain ourselves with the further elaboration and nuances of this analog-digital difference (which is itself an awkward dualism!), it points to a tension and potential conflict within the very notion of trace and traceification, which sits at the analog-digital threshold as a point of conversion. As we argue in Chapter 2, this also means that the distinction between trace and data is fluid and perspectival depending on how each is theorized, expansively or reductively. We also suggest that traceification and datafication processes bear a dialectical relation to one another; datafication begins with traceification, but data can in turn become traceified. In addition, the connectivities and relationalities that enable tracking, discussed further in Chapters 3 and 4, are not simply the technical pipework of the economy of traces—digital or analog—but also cultural reflections of what is and is not to be made observable via processes of traceification. While it is generally accepted that the "digital," commonly understood as technology, cannot be readily parsed from the social (Orlikowski and Scott, 2008), the digital and social are being continuously entwined in new ways to generate cultures of marking objects for tracking and accounting.

In summary, this book intends an approach to the analysis of traceability that neither ignores advances in digital technologies nor is hostage to their specific manifestations. Such an approach, if successful, can avoid the dangers of digital determinism, on the one hand, and cultural reductionism, on the other.[1] It also enables the navigation and expression of commonalities across the different value-purposes, traceifications, forms of tracking, and their accounts, despite seemingly very different practices, devices, and concerns—from financial regulation, to supply chain management, to self-tracking and many other settings. Scholars of information systems, computer science, and social scientists more generally have addressed issues about the nature and impact of digitalization for new markets and organizational forms, such as platforms. In particular, there is much debate about the gig economy and new forms of labour discipline and surveillance.[2] This book will touch on these debates at various junctures, but it is only and primarily concerned with their relevance for traceability as an implicit value or logic of marking, organizing, and accounting for ourselves, both individually and collectively.

[1] This is not to underestimate how the supply of hi-tech innovation can generate demand (Gordon, 2003)

[2] For example, the following is taken from a vendor promotional website, "Employee tracking software that tells you exactly how your employees use their time. Get powerful data for a more productive and efficient team". https://www.insightful.io/employee-monitoring (accessed 27 November 2024).

Traceability as Value and Purpose

The notion of "traceability" has two aspects. On the one hand, it embraces capabilities for the production of traces (trace-*ability*) as traceification, which enables objects, people, and processes to be observed and tracked through space and time. On the other hand, it connotes the evidentiary nature of traces and their connectedness (*trace*-ability) to phenomena of interest. The first sense is generative, the second is forensic, but they are interwoven and point to "a system that makes it possible, in real time and retroactively, to predict and record reliably the journey of an object from its origin to its destination" (Muirhead and Porter, 2019, p. 425). Beyond this general conceptualization, traceability as the production and use of traces is valued for different reasons, can be organized according to different value-purposes, takes many different practice forms, and has a long history, as long indeed as mankind and human communities have hunted animals and cultivated crops.

Traceability can be understood as an idea or aspiration, as an imaginary of control (e.g. Ortiz, 2022), and as something that is promised by politics, organizations, and systems of thought. It is proposed that individuals, organizations, and states seek to operationalize this imaginary of traceability according to three ideal-typical values, which are entangled with each other in practice but can be theorized as analytically distinct orientations towards *quality*, *security*, and *discovery*. This threefold model also suggests affinities between values and logics, and the manner in which they organize templates for practice and action (cf. Boltanski and Thevenot, 2006). Hence, our concept of "value-purpose" can be taken as more or less synonymous with that of "logic" in this sense.[3] And having initially articulated traceability in practice in terms of traceification, tracking, and accounting, we can also approach it from the "top down" as an idea, value, and aspiration that is mobilized for quality assurance, for security purposes, and in the discovery of new facts and bodies of knowledge. Let us consider each of these values-purposes or logics in turn.

Quality

Preoccupations with tracing for quality-assurance purposes have a long history within agricultural and related product markets. Buyers would often directly sample goods before committing to purchase, but knowing their origins was also important, hence the importance of labels, tags, and other certifying devices and

[3] Borrowing from Habermas (2015), we might think of value-purposes not only as logics but also as "traceability-constitutive interests".

traces created by producers. For example, the classification and labelling system for French wines emerged to regulate quality and to enable consumers to know where and when the wine in a specific bottle was produced, together with other features of its composition. In settings like this, a label or tag moves physically with the product as a trace of quality and enables a product to display its identity and origins. The production of such labels is often regulated, and there may be communities of experts and inspectors who engage in practices of tracing and auditing to ensure that the institutional "promise" of origin and identity indicated by a label is worthy of trust (Fourcade, 2012).

As sustainability-oriented change agendas have gained momentum, contemporary consumers want to know more not only about the functional quality and nature of products but also *how* they were made and by whom. The logic of quality is therefore elastic and has expanded in scope to include, for example, the traceability of environmental and ethical values in supply-chain practices. Importantly, advances in digital automation have transformed the capability for traceability, in the sense of trace production, in many areas. For example, GPS systems can now capture and record the exact point of harvest and trawler details in the setting of fishing (Lewis and Boyle, 2017). These traces are not the thing itself, or even a representation of it; they are rather the "digital double" of the specific catch and reference it. Digitally constructed traces of origins can also take a material form as a tag and become reattached to the real object (as bar codes have done since the 1970s), enabling the tracking of the object as it moves. The tagged fish carries with it its own mini-story of where it came from, as an "account" that is valued in the marketplace—provided, much like a wine label, that it can be trusted. In these settings, the traceability provided by tags is a promise but not a guarantee of quality, in part because the very idea of quality now includes the ethics of production and also because fraud is still possible and systems may fail. Hence, the evidentiary face of traceability in the form of inspection, auditing, and evaluation practices has emerged to test and demonstrate the trustworthiness of that promise on behalf of those who rely on it.

Beyond the setting of foodstuffs, traceability as the production and use of traces is important to manufacturing in general. Batch identifiers, product tagging, and barcodes (Kjellberg, Hagberg, and Cochoy 2019) are all critical tracing technologies that support transactional efficiency, enable management control, and are a source of assurance to the end consumer. For example, traceability for quality control is evident in the pharmaceutical industry, where companies must have a demonstrable capability to identify when and where drugs were produced so that instances of defective production can be tracked back to their origin. Relatedly, in their study of the formation of a market infrastructure for legalized cannabis, Pflueger et al. (2019) show how an accounting system designed by the regulator was important in defining the boundaries of the market and in enabling the tracking of cannabis from individual plant to consumer, for both quality and security reasons. Lezaun (2006) analyses similar systems for precise life-cycle control and traceability in the context of genetically modified (GM) foods. These examples suggest how the organized production and use of traces for quality assurance constitute an "infrastructure of traceability" (Power, 2019). They also suggest that the objects subject to traceability for quality are not always well defined and can be especially elusive when they are substances that undergo transformation and mixing in production chains. The promise of the quality of such production processes may only be redeemable by specialist analysis to *discover* the presence or non-presence of, say, a controversial chemical substance. Hence, values of quality and discovery overlap in many industrial settings.

Security

The second value-purpose or logic of traceability is that of security in its broadest sense. It concerns the production of knowledge oriented to the control and predictability of people and things, understood implicitly or explicitly as risks to be managed. States have been interested in knowing the characteristics of populations since the emergence of statistics (Hacking, 1990), so much so that state-building and technologies of population tracking and counting have often co-emerged, characterized famously by Foucault under his notion of bio-power (Foucault, 2008). For example, passport systems (Torpey, 2018) can be thought of in terms of traceability in our dual sense. On the one hand, they involve elaborate processes to control the production of passports as official traces of a person and proof of their identity as citizens. On the other hand, they involve equally developed practices of inspection and analysis of passports at national borders. Arrangements for tracking people may also vary greatly due to different cultural and value orientations, not least in relation to values of freedom and human rights. In the United Kingdom, internal identity cards have often been proposed but rejected; in the United States, the driving licence plays a de facto role in identity validation (Whitley and Hosein, 2008).

Security in its broadest sense may also include the health of a nation, which underpins economic performance. The onset of the pandemic in 2020 made this painfully evident as efforts to trace specific carriers of the virus in its early incubation stages could not control the spread, which eventually required the isolation of general populations and the restriction of movement. In anticipation of the easing of this lockdown, the United Kingdom initiated a system for "track and trace" by which individuals could be informed if they had contact with a carrier and were then obliged to isolate. Many countries invested in similar digital traceability practices, which had to be assembled in haste from many different elements. Political cultures of enforcement varied considerably, and there were failures and unintended consequences, but a COVID "passport" was eventually created to allow fully tested individuals to move more freely across borders. In short, in times of emergency and threats to security, the traceability of persons can become a pressing concern involving the rapid creation of technology solutions.

The manner in which traceability is organized according to values of security will vary according to political orders. We know that more centralized and totalitarian states invest heavily in technologies of granular population observation and control. For example, in the former East Germany (German Democratic Republic), the Stasi sought to know every detail of individuals' lives and created a population of internal spies to watch and create traces of them in files (Funder, 2002). Traceability as capacity for surveillance like this has undoubtedly been supercharged by the advent of digital technologies, including facial recognition software, which now enables all manner of individual behaviour and movement to be tracked and assembled into evaluative accounts, such as whether someone is a "good citizen" or not and deserving of a specific grade for social credit (Bach, 2020; Xu et al., 2023). Scholars have written extensively for many years about the newly intense forms of surveillance that digital technology now enables and which are common to predicting both consumer behaviour and also the seeming riskiness of citizens to the state and social order (e.g. Ericson and Haggerty, 2006). It is therefore evident that the organization of traceability according to security values, whether by states or corporations, is inherently political and often controversial, involving risks to values that many regard as fundamental, such as privacy and freedom of speech (Veliz, 2023; Zuboff, 2022). Furthermore, technology has itself created new opportunities for crime and insecurity (Van der Wahen and Pieters, 2015). Yet, it is also inherently difficult to isolate and regulate "bad" surveillance practices from the data infrastructures in which they are embedded.

Discovery

So far it has been proposed that traceability is valued for the generic purposes of assuring and controlling both quality and security. We might say that these value-purposes represent distinctive forms of knowing people, things and processes. Yet traceability, as the production of traces with evidentiary signification, also has a distinct value-purpose in the generation of the forms of explanatory and predictive knowledge that we associate with idea of "science". This points to a "logic of discovery" not in Popper's (1959) sense of a method for proper science, but as a shorthand for the general organization of science as investigative and device-saturated practice oriented to producing new—and testing old—facts about the world. Scientific practice as this work of potential discovery can be understood in part as seeking to establish causal connections in the form of associations between traces of phenomena, as potential signs of things in the world, associations that may confirm or disconfirm prevailing theories. Traceability oriented to discovery in this sense is not a matter of individual breakthroughs and achievements but is infrastructural and collective in nature. It involves systems and instruments that are organized for the production of *traces* of phenomena, and the specific tests and experiments that transform these traces into *data*, and which ultimately produce *facts* about nature (Rheinberger, 2011). A prominent example of such an infrastructure for discovery is the CERN particle accelerator and its high-energy physics communities (Knorr-Cetina, 1999) (see Chapter 9).

Traceability for discovery can also be explicitly *forensic* in character. Here, scientific practice is routinely enrolled for the quality and security purposes of tracking people, viruses, and chemicals, including carbon emissions. Indeed, forensic science as the analysis of "traces" has become a decisive resource for criminal investigations (Wiltshire, 2019; Servida and Casey, 2019). Its purpose is to discover and explain the "cause of death" and who is likely to have committed a crime. Discovery in such settings depends on the attribution of evidential strength, via Bayesian analysis or similar, to the trace(s) based on laboratory testing. Practice has evolved and so too have the traces of human behaviour and identity that it routinely examines, from fingerprints (Coles, 2001) to DNA (Aronson, 2007). Indeed, the latter has significantly extended the time horizon of discovery, thereby allowing "cold" cases to be reopened and solved.

Traceability as discovery is not restricted to scientific practice as conventionally understood. For example, "forensic accounting" (Williams, 2013) is a form of investigatory practice that draws on a range of traces as clues to a puzzle, usually about the "location" of financial assets that may have been deliberately hidden. In effect, forensic accounting uses existing documents as fragments to creatively reconstruct as far as possible a trail, an "audit trail", which enables the location of stolen assets to be discovered and known. The skill and professional judgement of the investigating forensic accountant is needed because such trails as connections are always incomplete, either by criminal intent or by neglect. Digitization has also spawned the data sciences with capabilities for traceification and datafication on a grand scale, enabling the discovery/generation of new patterns and relations, often with predictive power. Hence, to repeat, we should not associate the logic of discovery only with mainstream scientific activity. More generally, many professional and managerial practices, not least risk management (Chapter 7) have their own bodies of domain knowledge through which they engage in forms of exploration, experimentation, analysis, and tracing to discover things relevant to their practices. This is not formal science but borrows its general mode of rationality and evidentiary processes (March, 2006). Even organizational routines can be sources of innovation and discovery in this sense (Deken et al., 2016).

Traceability Hybrids

These three ideal-typical value-purposes or logics that mobilize traceability practices are rarely found in their pure form. They are always entangled with one another in empirical settings. Take, for example, the efforts of states to combat financial crime and money laundering. Regulation in this area focuses on the critical risk point of the individual-organization interface, namely the interaction and transactions between potential criminals and the banking system. States leverage the commercial interest of banks in "knowing your customer" for security purposes, and banks have built elaborate infrastructures to comply with financial crime regulation. Traceability in this setting involves "discovery" of the identity of a transactor and the source of their funds. Routinely, passports are used as proof of identity, but there are also specialist organizations, such as Experian and Kroll, that trace people and their activities. The case of AML (anti-money-laundering) regulation is therefore a hybrid—all three value-purposes are present but at different levels and in different ways. *Security* can be understood as the overarching value. The *quality* of AML systems at the organization level is the primary focus for board regulatory oversight. And organizations have been required to invest in statistical methods and other domain-specific forms of knowledge to *discover* and generate the "risk profiles" of customers, and thereby to understand ex ante causes or drivers of money laundering. Hence, when we set out to analyse traceability as organized practice in action, we always encounter a complex of these ideal-typical value-purposes. Figure 1.1 provides a topological representation of the overlap between the different value-purposes of traceability, their objects of concern, and some indicative methods.

We need to remain mindful that these purposes as value-orientations, both individually and in combination, are always aspirational. They may even be fantasies of observability and connectivity, which are themselves dependent on imperfect artefacts. On the one hand, technologies for traceification can generate new objects of interest and accounting, and new action-potentials, such as the notions of a "carbon footprint" (Kornberger et al., 2017; Mennicken and Kornberger, 2021) or "impact" (Power, 2015). On the other hand, they can fail in their operationalization for many different reasons: the stated purpose may exceed technical capability; there may be fraud and error at source where tags and labels are wrongly coded; and there is always the general nature of objects, such as people, things, and processes, to be more than, to *overflow*, their traces.

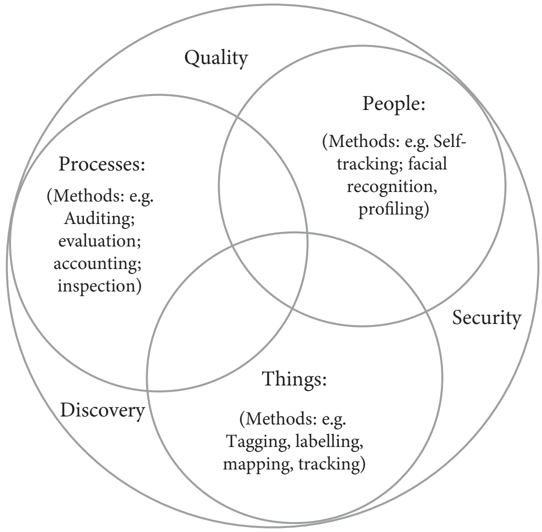

Figure 1.1 Traceability: value-purposes, objects, and methods.

Consequently, crises of traceability are normal and motivate new or intensified attention to value-purposes. The 2013 UK horsemeat scandal, in which consumers were misled about the composition of their food, led to widespread reform in foodstuff traceability and inspection (Elliot, 2014). Furthermore, people and things can be lost "without trace". Despite extensive investigative efforts to locate the final resting place of Malaysian Airlines MH370, and high-level assurances that it would be found, its whereabouts remain a mystery at the time of writing (to the general public at least). This event was especially shocking for a world in which digital technologies have increased expectations of perfect traceability and with it the belief that nothing can be lost. We accept that traces of ancient civilizations can disappear, but not a large modern aircraft subject to advanced tracking technologies by multiple aviation and military authorities.

Beyond these failures and scandals, it is also clear that traceability is not valued in the same way by all groups and individuals, especially regarding security, as already noted. Hence, there is an ongoing politics of *counter-traceability* involving conflict with other values, such as privacy and the "right to be forgotten". More generally, we must always locate the values of traceability in their cultural context as specific modes of institutionalized memorizing and tracing, which are often implicit and also involve modes of forgetting through trace erasure and non-production. These questions of value, about what we trace and why we preserve some traces and not others, are also being overwhelmed by advances in digital automation that operate behind the backs of individuals, effectively "extracting" their data without their consent (Zuboff, 2015, 2019). While we return to this theme in Chapter 8, it is interesting to note in passing that European legislation to protect privacy—General Data Protection Regulation (GDPR)—requires organizations to demonstrate their compliance with it, and therefore to make the practice of "making people untraceable" itself traceable. In other words, practices to manage and reduce the traceability of persons ironically increase the demand for traceability of the processes that demonstrate it.

In summary, we can think of traceability at an ideational level as a widely diffused aspiration organized around three general value-purposes or orientations, namely quality, security, and discovery. If we return to the example of the marathon runner wearing a smart watch, then his self-tracking can be said to be mobilized by interconnected values of quality (running performance), security (attention to key health factors), and discovery (analytics and data production). Furthermore, the example suggests how traceability has become big business, with new markets for personalized health knowledge and much else. Demands for governance and risk management (e.g. food-quality assurance and sustainability) have also emerged and have been supercharged by the promises of digital traceability (Muirhead and Porter, 2019; Thylstrup et al., 2022). Indeed, the sheer variety of tracing and tracking practice is intimidating for the analyst—threatening to be so broad and seemingly ubiquitous as to make the very category of traceability impossible to grasp and determine. Yet the proposal of these three ideal-typical value-purposes provides our first movement towards theorizing the organization of traceability. The second movement requires that we turn to accounting.

The Accounts We Live By

Accounting is conceptually and linguistically related to accountability as a value in many societies. Accountability can be achieved in many different ways (Donia, 2025), but in one basic form, the idea of accountability is a relational triad in which A has the capability and is somehow obligated or compelled to give an account B to, or on behalf of, someone else C. Such a formal relational structure encompasses enormous potential empirical and theoretical variety and breadth. How accountability as this relational triad manifests itself depends on cultural context and the precise referents for, and relative focus on, placeholders A, B, and C. Are we primarily interested in the duties of the "givers" of accounts and the AB dyad; or are we interested mainly in the receivers or so-called stakeholders and users of accounts defined by the dyad BC? Is the ABC structure fixed by legal or regulatory contract, or is it more a matter of interpersonal expectations? It is also often assumed that A and C are well defined entities and agents, and that the accountability problem is "simply" one of designing optimal forms of B to allow C to sufficiently observe and incentivize A. Yet, what if the accounting B has agentic, symbolic, and expressive properties of its own, even to the point of having the power to shape the identities and motivations of both A and C? Furthermore, what could "self-accounting" mean if A and C are the "same" embodied human being? At its most fundamental, A may stand for an "I" or self, which is shaped in the accounting encounter with an "other" C. The narration of self in the face of the other is a primordial form of accountability, in which A and C are reflectively co-formed but one that is shaped by prior norms and power (Butler, 2005, pp. 3–40). These and many other questions suggest that the relational space of account-giving is far broader than standard models of accountability and that methodological and normative choices must be made about the kinds of relations and the kinds of accounts that matter. Yet, one general claim can be made at this point: Bs are usually produced to make aspects of As observable and traceable for Cs. Yet, there is also always opacity, and traceability and transparency are always partial (Roberts, 2009). As Butler (2005, p. 43) puts it, "life might be understood as precisely that which exceeds any account we may try to give of it". Elsewhere in this book it is argued that we need to invert the traditional idea that accounting precedes auditing and tracking, an argument that forms a building block for a traceological theory of accounting. For now, we conceive of accountability and accounting very broadly (Vollmer, 2019), a breadth that is risky but is necessary to grasp both the meaning of the "accounts that we live by" and what is at stake. We consider three themes which underpin this breadth: account-abilities; worlds of accounts; and accounting agency.

Account-abilities

Levinas (1986) points to the uncanny nature of the encounter with the face of another as the primary source of human responsibility. In other words, the accountability structure ABC begins fundamentally with A's *recognition* of, and obligation towards, C (see also Butler, 2005, ch. 1). We might take this notion further in a more "phenomenological" direction by viewing the everyday accountings between individuals as the fundamental building blocks by which the social world is continuously sustained, reproduced, or changed (Berger and Luckman, 1966). The explicit production of accounts in the case of interpretations and explanations of individual behaviour is not normally required in everyday interaction, but the possibility of such production is tacit and underwrites all talking and interacting (Garfinkel, 1984a; Vollmer, 2019). Scott and

Lyman (1968) argue that accounts can be understood primarily and narrowly as forms of excuse and justification, which are produced by human agents when their behaviour somehow deviates from tacitly accepted norms. This influential conception of accounting as justification by exception builds on Garfinkel's famous "breach experiments" in which his students were instructed to violate norms in the home and with friends. Their puzzling behaviour triggered a mixture of reactions, including demands for "accounts," which revealed the unsaid norms governing expected interaction in specific situations. In short, an account must be given to justify non-conforming behaviour which disrupts tacit expectations, and A is deemed to have account-*ability* to do this and to provide an answer that C can accept.

Relatedly, but in a more general form, Habermas (1979) proposes that all talk is premised on a number of "validity claims" or presumptions that are assumed by others to hold, but which in principle can be "redeemed" or justified under challenge. In other words, the ability or competence for explicit accounting in Garfinkel's sense is generally inherent in, and a condition of, all human interaction and discourse. Indeed, Habermas wants to argue more strongly that such communicative accounting is a "transcendental" condition of the possibility of rational argument, conditions that point to ideal conditions of speech as a model of liberal democracy. While this notion of ideal speech is controversial, not least for its universalism (e.g. Hesse, 1995), it is nevertheless established by scholars since Garfinkel that human interaction is subject to continuous and more or less conscious monitoring or auditing of verbal and non-verbal cues. For example, such everyday monitoring determines whether it can be presumed that someone is being honest, or whether explicit accounts are needed to examine, justify, or challenge them. But, contra Habermas, the accounts we potentially live by and which inhere in communicative interaction should not be imagined as merely cognitive or free from distortion. As critical theorists like Butler (2005) remind us, accounts of and by selves are never cognitively innocent. The ABC accountability triads are themselves the products and expressions of institutionalized power structures and ideological forms, most importantly in the coding of the distinction between what gets made explicit and accounted for and what is silenced and without voice. Indeed, for theorists like Butler, traces become constitutively important to accounts in two senses. Not only are they the primary experiential matter—the verbal and bodily expressions to which we are constantly reacting and evaluating in the background—which make up continuous ongoing accounting as expectation-confirmation and sense-making. They are also the suppressed indications and signs of alternative accounts that could be given, but cannot find expression or must remain tacit (Vollmer, 2019). Critical accounting scholars have extended such arguments to the formal organizational level (Hines, 1988).

All accounting—of self and organization—is value-laden in the sense that it rests on choices and power relations, by design or otherwise, about what to make and not make explicit and "account-able", and therefore about the forms of accounting that count. From this general point of view, human everyday lives are necessarily permeated by account-abilities in the sense of capacities for narration of self through which relations to others are constituted. It may be argued that blurring the distinction between accounting and everyday sense-making like this is going too far. Nevertheless, studies of practice also demonstrate the deep continuities and exchanges between so-called everyday and organizational accounts as actors strive hard to generate "situated functionality" from arrays of activities and interactions (Ahrens and Chapman, 2007). These continuities and connections between the everyday and organizational are becoming even more evident under conditions of advanced digitization (Alaimo and Kallinikos, 2024, p. 101). In sum, the formal accountability triad ABC is in reality an accounting assemblage that blurs the boundaries of the personal and organizational by connecting "the production of accounts and accountabilities in everyday life with the more technical aspects of professional practice in a continuum" (Vollmer, 2019, p. 29).

A World of Accounts

The second way in which we must broaden our understanding of what counts as accounting is empirical, meaning simply that there is now a large and growing volume and variety of accounts that pervade both organizational and personal life (Vormbusch, 2022). In addition to the conventional forms of financial accounting, budgeting, costing, and cash-flow statements and their underlying books and records, there has been an explosion of metrics, both financial and non-financial, which seek to capture and measure a wide range of phenomenon, such as employee satisfaction, intellectual capital value, and other possible "drivers" of, and risks to, financial performance (Bukh et al., 2001). Indeed, studies of accounting grounded in information economics focus less on the conventional elements of financial accounting and more on the signalling properties of information and disclosure in general, thus also considerably broadening the horizon of what counts as accounting. As customer retention has become critical to many businesses, measures of retention risk have also been developed, and risk management more generally has moved towards the production of lead indicators as an accounting ideal (see Chapter 7). In addition, organizations now account and report on a large number of activities under the broad label of sustainability: measures of workforce diversity, carbon emissions data, and much else have become commonplace, although problems of reliability and comparability are also evident (Kotsantonis and Serafeim, 2019). These relatively new accountings are often hybrids of the numerical, narrational, and visual, organized into "dashboards" that connect different sub-accounts and enable the tracking of performance (Pollock and D'Adderio, 2012; Quattrone, 2017; Ronzani and Gatzweiler, 2022). Indeed, while accounts have always had visual properties, visualization has become a form of accounting in and of itself (Quattrone et al., 2021).

Many other practices, which exist beyond formal organizational boundaries, also produce what must count as "accounts", such as audit reports, rankings, compliance certificates, credit ratings, social media ratings, social credit scores, risk maps, and much more, including even the humble list (Gawande, 2010; Crvelin and Lohlein, 2022). In addition, employees are also known to keep their own idiosyncratic and local vernacular accounts (Kilfoyle et al., 2013), including "black books" and other devices, which supplement or replace formal systems. And, as the example of the smart watch with which we began this chapter shows, technologies and sensing devices of many kinds mediate and enable possibilities for accounts of personal bodily "performance"—athletic, health, and emotional. With such accountings we move beyond a purely justificational or evidentiary conception of accounting and accountability. Rather, the accounts produced by new capabilities of tracking are implicated in the creation of novel objects of concern and new account-*abilities* at a high level of granularity for individuals and organizations. They are generative of new worlds of practice and experience (Kornberger et al., 2017). Whether a new kind of self is constituted by such tracking-based accounts is a question we address in Chapter 8, but the digital transformation of formal organizational accounting is undoubtedly blurring traditional boundaries and temporalities (Bhimani, 2015; Bhimani and Bromwich, 2009; Knudsen, 2020), as well as shaping the production of accounts to suit the application of coding (Rowbotham et al., 2021). One further conspicuous effect of these transformations is the crossover between work and play, which is evident in the emergent gamification aspects of performance management and accounting.

Despite all these arguments the differences between "everyday" accountabilities and highly institutionalized "accountings" may still seem obvious and great. They certainly involve very diverse bodies of knowledge, expertise, expectations, and settings. So it might seem that the very concept of "accounting" is being stretched to the breaking point and is in danger of becoming meaningless. Yet, as we shall see in Chapter 8, this expansion of accounting is not an analytical contrivance. It corresponds to something real to be explained theoretically and empirically. Devices are expanding their reach to make the self and its "everyday" nature in Garfinkel's sense newly visible and calculable (Fauré et al., 2010). Programmed mechanisms of feedback multiply, which are more than simply sense-making and which challenge the supposed interiority of sensed experience (Vormbusch, 2022). Hence, the "quantified self" (Lupton, 2016c) represents a digitized and mediated intensification of Garfinkel's notion of everyday accounting, reducing the supposed difference between everyday and institutional accounting as the behavioural reach of quantification is extended (Thaler, 2016). Indeed, traditionally conceived differences between institutional and personal worlds are becoming harder to sustain (Vollmer, 2019). Boundaries are frequently crossed in both directions and continuously remake each other via the fluidity of data in which organizations and personhood are becoming more distributed and decentred (Alaimo and Kallinikos, 2024). Limiting what counts as accounting to institutionalized and specialized practices, say for insurance companies, would arbitrarily restrict our analysis of traceability and accounts production.

Latour (2005) also provides support for this expansive notion of accounting. He argues that the very agency of things and people can only be grasped and accounted for to the extent that they leave observable traces that can be tracked and assembled into accounts (2005, p. 53). These traces in his sense are not only "out there" as products or imprints of both human and non-human agents, but can also be produced explicitly by the analyst-sociologist or critical observer who must "traceify" objects and track their connections as an activity of accounts production. In this respect, there are methodological affinities between Latour's thinking and studies that attend closely to the traces of actions in order to map and analyse the grammar of organizational routines (Pentland and Rueter, 1994). A good social scientific account for Latour is one that tracks a network of connections of interest, and which demonstrates the forms of connectedness that are productive of our singular and perhaps mythical conceptions of agency. This analytic work involves the effortful re-presentation and mapping of the traces left behind by these various agencies. The result is what he calls "risky accounts" which can "easily fail" and often do (2005, p. 132). Importantly, for Latour, there is no difference in principle between this accounting work as it is done by both analytic and non-analytic actors, both of whom must become "attentive to the material traceability of immutable mobiles" (2005, p. 226). Indeed, he suggests that traceability in this sense of network construction underpins all accounts production as the connecting of traces created by people and things.

If we dare to follow both Latour and Garfinkel, accounting emerges as an important general model of knowledge production and experience. It is a continuous thread that connects and mediates worlds of everyday interaction and specialist analytical practices, like the natural sciences or insurance. Using a predigital vocabulary, Latour argues that we (meaning anyone claiming to do the sciences of the social) have to put these "humble paper tools in the foreground"

(2005, p. 230) and attend to the circulation and connectedness of traceable documents in all their varied forms. Indeed, for him, this connectedness as traceability, and the accounts we generate from it, is the very source of what we mean by "society".

Latour (1986, 2005) is not especially precise about the ontological nature of traces, although he inspires us to take them seriously as features of fact-building. He tends to equate them with documents and inscriptions and sometimes things. Being concerned primarily with the deployment of durable traces to produce "hard" facts in agonistic and competitive scientific settings, he is also relatively silent about the variety of institutional forms of traceability and the specific circuits of connectivity, which enable tracking and accounting. He is primarily interested in the active construction and mobilization of the connections that underpin science, law, and other projects and their accounts, rather than the preoccupation in this book with how traceability in general is organized and may be changing. However, his level of analysis helps to cement the traceification-accounting relationship and reminds us that if the traces, and the processes by which they are traceified and datafied, shift systemically, then this will have consequences for the accounts that we produce and live by, all the way from the everyday to the institutional and back again.

Agentic Accounts

The third theme, which builds on the observed variety described in the previous sub-sections, concerns the many effects and roles that accounts may have, regardless of their formally stated purposes. In this respect, it can be proposed that accounts as the Bs of the ABC accountability relation are socio-material artefacts that may have their own agency and contingent conditions of operation. For example, it has been established that accounts may have many different roles and effects, which drive their technical elaboration and variety. In their seminal paper, Burchell et al. (1980) draw attention to the role of accounts beyond the mainstream conception of information for decision-making, a conception that, in the case of financial accounting, was "invented" as part of the post-war rise of decision sciences (Young, 2006). It is argued that accounts can be used as "ammunition" in adversarial settings (e.g. contested takeovers) and can provide ex post forms of rationalization and justification (much like the "excuses" of Scott and Lyman [1968]). As we discuss further in this book, accounts can be generative of their own facticity and seeming objectivity, and studies have also demonstrated their expressive and affective aspects (Chenhall et al., 2017), including their role as vehicles for supererogation (Horvath, 2023). Hence, accounts are not simply channels of information for well-defined decision points but, in specific contexts, produce or reduce comfort, and can alleviate or amplify fear and anxiety (Guénin-Paracini et al., 2014; Vollmer, 2019). Accounts are known to be flawed by practitioners who nevertheless use them as resources interactively to generate control activities (Ahrens and Chapman, 2007).

Quattrone (2015), like Garfinkel, argues for a *maeutic* and processual conception of accounts as forms of individual and collective sense-making that are not static, and in which reality is a continuously unfolding referent of practice where the tacit becomes explicit. Furthermore, accounts are necessarily and inherently projective and can be thought of as technologies and devices that stand in a relation of temporal co-formation with humans, meaning that the many different accounts that we live by are not simply modes of excuse, justification, and legitimation to be tested and proven by audits. They are themselves a constitutive part of the many and changing ways in which we narrate who we are and might be, collectively and individually (Brown, 2009). Indeed, even accounts in explicitly calculative form are generative narrations that bring together and organize traces and purposes and which project future possibilities (Stiegler, 1998; Matringe and Power, 2024). Hence, we must always be mindful of the dual temporality of all accounts, which is inherent in their traceification, of marking what has happened and projecting what will or should happen (see also Chapter 7). This duality is not just a new version of the age-old tension between backward-looking stewardship and forward-looking decision relevance. It is a more phenomenological point about accounts and accountability that "pull" the human user towards an open future in the very act of marking what has been (Chua et al., 2024). This dual temporality of accounts, which operates between their historical residues and imagined futures, is more than just a proposition about the changing nature of accounting forms; it requires us to position accounting within a process of on-going design, which seeks realization and operability, and for which continuous repair is normal (Bottausci et al., 2024).

In effect, accounts, in all their variety, should not be understood narrowly as passive conduits or intermediaries for information; they can be agentic mediators in their own right, and in this sense the accountability triad of ABC is a socio-material assemblage consisting of three agent-placeholders, two human (so far, but this is changing) and one non-human: giver, receiver, and the accounts themselves, which enable aspects of the giver's actions to be "tracked" by the receiver. We should also be mindful of the institutionalized mythologies and fictions that pervade this triad, not least the idea of the "user". And we should be wary of overstating the agency of accounts; they are always, even in the most automated settings, embedded in human practices. It is also well known that: organizations produce more information than they use (the ABC triad is full of surplus and overflow); accounts are often produced for symbolic reasons (March, 1987); and even flesh-and-blood users are somehow "made up" by the accounts through which they engage in organizational life. These messier agentic properties of accounts are not always evident from their official rationales.

In summary, we have considered three themes which suggest that when we think of the different kinds of accounts that organize our lives and behaviour, as individuals and organizations, then we must not prematurely circumscribe what counts as accounting and must a priori take a broad view of accounts as stories, often told with and through numbers. The accounts we live by can be broadly defined as the diverse forms of narration, numeric representation, and summation that organize and mediate human and organizational relations. So we should not allow only professional accountants to define and bound what is, and is not, the apparent nature and limits of accounting. The formal, specific, collective forms of accounting that make up the pedagogy of accounting students as aspiring practitioners are themselves specialist abstractions from, and extensions of, the primary interactive settings that interested Garfinkel and his followers. The history of accounting demonstrates this. For example, in feudal Britain, stewards entrusted with the management of the affairs of a manor were required to present in person accounts indicating their "charges and discharges"; and early modern merchants in northern Italy used double-entry bookkeeping to record and control transacting and indebtedness. In both these historical contexts, accounting was not understood abstractly but was deeply embedded in social relations in which personal credibility and reputation were always at stake and where justification might be required (Matringe and Power, 2024). Today, highly institutionalized and rationalized practices of accounting seem to have a more self-referential and dis-embedded life of their own within formal organizations and markets (Vollmer, 2007). This is no doubt a product of the professional mobilization of accounting practitioners and their institutionalized power to define knowledge. But accounting numbers also do not speak for themselves, and an important focus in the project of critical accounting has been to reveal and deconstruct this seeming autonomy, to expose the embedded nature of accounting practices and the interests that mobilize them, and to "bring society back in". This critical agenda extends to the production of alternative or "counter accounts" that reconnect the everyday and its values to organizations more explicitly. The sustainability accounting agenda that has taken off in the 2020s (although elements of it reach back decades) represents both the expansion and potential normalization of this critical agenda, although, at the time of writing, these efforts are still characterized by considerable fluidity, experimentation, and political contestation.

To conclude: the notion of the "accounts we live by" has been introduced as a provoking category. It is intentionally jarring and dissonant in order to challenge conventional ways of thinking by practitioners and scholars about the technical boundaries of accounting. It is also implicitly expressive of the constitutive power of accounts for human experience. In this sense, accounts can be understood most generally as products of the restless dialectic between life and form that Simmel (1971 [1918]) articulated, a dialectic in which form continuously dominates but cannot escape its grounding in life. In later chapters of this book, it is argued that the accounts we live by are undergoing a systematic shift, a shift that is symptomatic of changes in this life/form duality itself.

Chapter Summaries

Scholars have studied the co-constitution of accounting, organizing, and society for at least four decades and have shown how accounting systems are contingently influenced by organizational and institutional factors (Chapman et al., 2009; Miller and Power, 2013). They have shown how accounting technologies and practices have agentic properties of their own, are carriers of values, and shape the attention and behaviour of organizational actors, even to the point of generating new organizational realities. There has also been increased attention to the raw material from which such accounts are fabricated, including the inscriptions and material technologies, and the transactional systems and rules, which underpin accounts production. Yet the traces and data on which accounts are grounded, and which have an institutional life and significance of their own, deserve more theoretical and empirical visibility in their own right (Alaimo and Kallinikos, 2024). This book is an invitation to see how the changing production of traces and the accounts we live by are intertwined in the organization of traceability. The scope of the argument is broader than the formal accounts produced by organizations for capital markets or for internal-control purposes, and includes the many narratives that we produce of ourselves and others as we navigate and mediate between personal, social, and organizational life. This breadth is not simply a methodological contrivance. It reflects the fact that there are increasing exchanges between the personal and organizational as digital transformations reconfigure and re-proximate these relations.

Each of the chapters that follows discusses the organization of traceability and its components from a different thematic angle. They include a number of vignettes for illustrative purposes, which are drawn mainly from the author's experience and do not reflect systematic fieldwork. In addition, each chapter concludes with a series of summative propositions that capture the essence of what has been said and are provided to aid reader navigation. In broad terms, Chapters 2–5 build key concepts and Chapters 6–10 apply them to contexts. However, the unifying aspiration is that all these chapters in aggregate provide a synthetic grasp of traceability from different entry points. They can also be understood as building blocks for theorizing the traceological basis of accounting and how it is changing in the "new" economy of traces in the digital age.

Chapter 2 deepens our introductory remarks on traces, traceification, and tracking. In doing so, it unpacks the relation between traceification and datafication, and develops the distinction between tracking-led accounts and accounts-led tracking, which is at the centre of the book's thesis regarding a shift in the accounts we live by. Drawing on the analysis of traceability and accounts in Chapter 1, the idea of the "traceability complex" is advanced in which value-purposes, traceification, datafication, tracking, and accounts are interconnected.[4] The idea of a traceability complex provides a working template for a traceological approach to accounting and governance.

Chapter 3 revisits critical studies in accounting to suggest that, despite the Latourian-inspired "inscriptivist" turn in accounting theory, a primary focus on quantification and its effects has tended to understate the constitutive role of traces and audit trails. We invert the traditional conception that accounting comes before traceability and auditing, thereby placing the audit trail, as idea and practice, at the centre of our analysis. Furthermore, a core pragmatist proposition in this traceological approach to accounts production is that *tracking precedes representation*. The chapter also develops the notion of "primary traceification" as a boundary condition of the audit trail that is value-laden in the sense of making some things visible and accountable and not others. Chapter 4 provides further development of the audit trail concept, emphasizing its role as mode of connectivity in general, which must be generated to make tracking and accounting possible. It is argued that infrastructures of connectedness and practices of connecting co-develop. The argument then turns to variations in the nature of audit trail connectedness under conditions of digitalization. The thesis is that audit trails become less deep and less dense, and this is associated with the rise of tracking-led accounts. The chapter concludes by articulating the paradox of audit trail connectivity: as digital connectivity increases, the audit trail as a basis for human oversight decreases and may disappear entirely.

Chapter 5 approaches traceability as a core feature of *organizing and governing*. It builds on prior studies to argue that traceification and tracking are

[4] It has been pointed out to me that the term "traceability complex" might refer to a psychological form of complex, i.e. a condition of fearing or even being obsessed with being traced or engaging in tracing activity. Although this is not the intended theoretical meaning of the concept in this book, it is not entirely without salience. For example, as we see in Chapter 8, widespread enthusiasm for self-tracking could be interpreted as a "traceability complex" in this sense. Furthermore, to the extent that psychological concepts can be extended to the societal level, the economy of traces potentially reflects a collective "complex" about traceability. For these reasons, while the notion of complex will be used very differently, this ambiguity is also useful to retain.

constitutive of the entities we designate as organizations. Borrowing from theories of documentality and scripting, it is argued that organizations are constituted by scripts that may be more or less explicit and which provide a point of orientation for communication, both formal and informal. It is also proposed that the stability of organizationality over time and space requires some minimal degree of self-observation or oversight, often enabled by accounting systems. The chapter develops the distinction between tracking and "oversight", arguing that they are always combined but may vary in mix. Oversight is conceptualized as layered, drawing on metaphors-in-use of "lines of defence" and "supply chains". Oversight practices are generally enabled by audit trail modes of connectivity and hence systemic changes in the latter will have consequences for the former and the governance supply chain. Chapter 6 builds directly on Chapter 5 and uses the traceability complex template to compare several governance and regulation settings. It is argued that under conditions of digitalization, conventional notions of governance are being de-layered and platformized. The "traceability lost" thesis is developed by an analysis of the debate about the audit and governance of algorithms. It is argued that our capacity for trace-generation at scale is overwhelming the human evidentiary basis of traceability.

Chapter 7 draws on Stiegler and others to establish the ontological relations between traces and temporality. It builds on this argument to develop the idea that anything can be coded to be a trace of the future, which can in principle be tracked and accounted for. It also revisits Hilgartner's theory of risk-object formation to show how the temporal depth of such accounting is critical and controversial, and influences credible risk-indicator design. In general, the idea and practice of risk management is positioned as a distinctive general form of traceability complex that embodies contrasting temporal orientations to past, present, and future. Digital forms of traceability can also be framed as risks—in aiming at the elimination of uncertainty and at predictability, they potentially elevate a continuous algorithmically produced present over human projective capacities. Chapter 8 turns its gaze towards the humans who are entangled in different traceability complexes. Building on relational theories of the self, the chapter first revisits studies of the formative role of quantification and accounts, arguing that in digitized self-tracking, this takes an extreme form, even to the extent of such tracking becoming a valued end it itself. Three critical vectors of self-formation are considered: profile construction and curation; the self as feedback machine; self-tracking and data-doubles. In each case, the traceification of self involves exchanges across the boundaries of organizational and private life, which blur and deconstruct those boundaries. And with this blurring, the very notion of performance has expanded its scope and reach. In sum, the chapter proposes that traceability complexes are increasingly populated by trace-enabled humans within thin, digitally mediated relationalities, a transformation process that can be theorized as cyborgization.

Chapter 9 considers the implications of traceability as a distinctive social epistemology and revisits the value-purpose of discovery. The chapter discusses and distinguishes between traceability as practice and as method within relational theorizing of science. However, studies of big science also suggest a convergence between these two senses of traceability as data and knowledge production are increasingly platformized. Furthermore, a form of digital positivism is increasingly evident in both natural and social science in which digital data are more than a new resource but also define the nature of the investigatable empirical domain and thereby influence research agendas. To illustrate this strong thesis, it is argued that developments in accounting research represent a general form of this digital positivism—"accountics"—which is reshaping our conception of the empirical world and generating a convergence between the social sciences and managerial control. The chapter concludes by proposing the idea of a "critical traceology" focused on the mechanisms by which this digital positivism is expanding. Finally, Chapter 10 pulls these different arguments together to consolidate a traceological theory of accounting and to discuss how the digital economy of traces is transforming or intensifying audit society and expanding the range of human activities that can be thought of as "performing". If the audit society was essentially an oversight explosion, what we see now, sometimes designated as "surveillance capitalism", can be said to be a tracking explosion. The latter is changing the mix within organized forms of traceability from oversight to tracking.

Other than building a more traceologically sensitive theory of accounting, why does it matter to establish traceability as an object of analysis like this (cf. Muirhead and Porter, 2019; Power, 2019)? There are two main reasons. First, traceability complexes, understood as bundles of practices, are implicated in generating new markets for governance (Hinings et al., 2017), new forms of distributed agency in supply chains (Enfield, 2017; Bernstein, 2017; Kockelman, 2017), and new capabilities for surveillance by states and organizations (Zuboff, 2019). They also expand definitions of "performance" that span the organizational and the personal, and which, as we argue in Chapter 8, elevate "feedback" as a mode of evaluation. Traceability complexes are therefore material realizations of variable social demands to fix, memorize, track, and account for our individual and collective activities. They are sites of an emerging politics of traceability involving struggles over purposes, over the capture and extraction of digital data, over the ethics of tracking as observation, and over the manner in which the organization of traceability itself can be somehow governed, not least when it is saturated by algorithmic processes.

Second, as noted already, traceability is profoundly epistemic. This architecture of value-purposes, traces, tracking, and accounts is a way of knowing people, processes, and things, and of making them observable and thereby connected, both to others and to themselves. Philosophers through the ages have been preoccupied with seeing and observing as primordial forms of knowledge that underpin science and much else. As pragmatists, like Rorty (1980), have argued, this claim to insight about the foundations of knowledge has given philosophy its professional identity as epistemology (an identity which he challenges). In this book we argue that it is practices of tracking, and their documentary and digital form, which utilize traceification and which make certain styles of observation and connection, and therefore knowing, possible (Latour, 1986). They do so not with a transcendental foundation but as a form of pragmatist rootedness in the world of material trace potential. Humans tracked prey long before they tracked marathon runners on apps.

The epistemology of tracking is necessarily permeated by interests that make some aspects of the world observable but not others. As we go deeper into the epistemic character of tracking, and how it conditions and is conditioned by observation capacity, the practice landscape becomes more differentiated. On the one hand, it can be organized to be evidentiary, forensic, and justificatory, underpinning hierarchical forms of regulation and governance. This tends to be productive of accounting-led forms of tracking. On the other hand, it must also be understood as creating often surprising, novel realities and new relations between people and things, including new topologies of organizationality and ways of steering and controlling societies (Kornberger et al., 2017). Whether we like the social orders generated by tracking or not, we cannot but reflect on their trajectory and their implications for the accounts that we live by.

In this book, we theorize these politico-epistemic stakes of traceability and their organization in terms of a fundamental tension between two different ways of configuring the relations between value-purposes, traces, tracking, and accounting. These different ways are ideal-typical, and in reality they are blended, but they define contrasting modes of organizing traceability in terms of whether they are oversight or tracking-orientated. These oppositions are stated simplistically, but this is no mere play on words. They define the very axis on which societies may be turning in the digital age and lie at the heart of a postulated transition between the motifs of audit society and surveillance capitalism. Humans have always lived in an "economy of traces" of some kind. However, today the accounting potential embedded in the traces that we datafy and the data we traceify and our ability to track and narrate our own lives and possibilities, and those of others, are greatly expanding and changing. If the "accounts we live by" refers to the role of accounting in generating the facticity of that which is valued by humans, then nothing less may be at stake in the analysis of traceability than the nature of human experience and purpose itself.

Chapter 1 Summative Propositions

The following propositions provide four working definitions of key concepts that are used throughout the book and two key claims that are developed in later chapters:

1. **Traceability is a capacity both to generate traces and to use such generated traces for evidentiary purposes.**
2. **Traceification is the process by which traces are generated for specific practical purposes where traceability is an organizing interest.**
3. **Traceability is mobilized by one or more of three general value-purposes or logics: quality, security, and discovery.**
4. **The "accounts we live by" are broadly understood as the diverse forms of narration, numeric representation, and summation that organize and mediate human and organizational relations.**
5. **Tracking precedes representation.**
6. **For all their evident variety, the accounts we live by are undergoing a systematic shift in the digital age.**

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